

Statistical Nonparametric Mapping: Multivariate Permutation Tests for Location, Correlation, and Regression Problems in Neuroimaging*

Nathaniel E. Helwig

Assistant Professor of Psychology and Statistics
University of Minnesota (Twin Cities)



February 13, 2019

* Helwig, N. E. (2019). Statistical nonparametric mapping: multivariate permutation tests for location, correlation, and regression problems in neuroimaging. *WIREs Computational Statistics*. doi: 10.1002/wics.1457

Outline of Talk

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- Simulation Study

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- Background Theory
- Simulation Study

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- Background Theory
- Simulation Study

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- Future Directions

Brief Background

Permutation Tests are Popular in Neuroimaging

Rise of Statistical Nonparametric Mapping

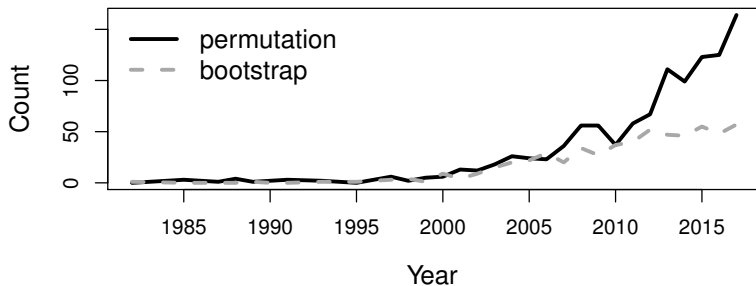


Figure 1: Timeline of the number of publications on permutation (black solid line) and bootstrap (gray dashed line) resampling methods for neuroimaging data.

History of Permutation Tests in Neuroimaging

Blair and Karniski (1994) and Karniski et al. (1994) had the first fundamental works on permutation tests for neuroimaging data.

- Applying the theoretical ideas discussed by Blair et al. (1994)

Holmes et al. (1996) solidified and extended these ideas to PET data.

- Arndt et al. (1996) independently proposed these ideas

Statistical nonParametric Mapping (SnPM) software (Nichols et al., 1996) is very popular (and still under development) to this day.

Eklund et al. (2016) demonstrated the utility of nonparametric inference.

Classic Permutation Tests are Old News...

Nonparametric inference via randomization and permutation tests has been around since Fisher's (1935) *The Design of Experiments* (see Chapter III).

Pitman formalized the idea to the two (independent) samples location test (Pitman, 1937a) and the correlation test (Pitman, 1937b).

Several authors have written on regression permutation tests (e.g. Draper and Stoneman, 1966; Freedman and Lane, 1983).

...But (Lesser Known) Modern Improvements Exist

Romano (1990) and Janssen (1997) prove one and two sample permutation tests provide asymptotically correct results for skewed distributions.

Chung and Romano (2016) propose robust two sample permutation test using modified Wilcoxon statistic (Wilcoxon, 1945; Mann and Whitney, 1947).

DiCiccio and Romano (2017) propose robust permutation tests for correlation and regression problems—leveraging the ideas of White (1980).

See Helwig (2019) for a discussion of how these ideas can be used to improve current implementations of permutation tests in neuroimaging.

One Sample (Paired Samples) Location Tests

Motivating Problem

Let A_{iv} and B_{iv} denote some brain signal recorded at the v -th voxel ($v = 1, \dots, V$) from the i -th subject ($i = 1, \dots, n$) during conditions A and B.

Consider the following model separately at each voxel

$$Y_{iv} = \mu_v + \epsilon_{iv} \quad (1)$$

where

- $Y_{iv} = A_{iv} - B_{iv}$ is the i -th subject's difference score at the v -th voxel
- μ_v is an unknown mean parameter for the v -th voxel
- ϵ_{iv} are iid error terms with mean zero and variance σ_v^2

Want to test **local null hypothesis** separately at each voxel $H_0^v : \mu_v = 0$ and the **global null hypothesis** across all V voxels $H_0 : (\forall v)(\mu_v = 0)$.

Test Statistics

For $H_0^v : \mu_v = 0$, we can use Student's (1908) one-sample t test statistic

$$T_v = \frac{\hat{\mu}_v}{\hat{\sigma}_v / \sqrt{n}} \quad (2)$$

where $\hat{\mu}_v = \frac{1}{n} \sum_{i=1}^n Y_{iv}$ and $\hat{\sigma}_v^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_{iv} - \hat{\mu}_v)^2$.

For $H_0 : (\forall v)(\mu_v = 0)$, use the maximum test statistic

$$T_{\max} = \max_v T_v \quad (3)$$

where the maximum is taken across all V voxels.

Permutation Inference

Randomization Assumption: Data from conditions A and B are exchangeable, i.e., $(A_i, B_i) \stackrel{d}{=} (B_i, A_i)$ where $A_i = (A_{i1}, \dots, A_{iV})$ and $B_i = (B_{i1}, \dots, B_{iV})$.

Permutation Lemma: There are 2^n possible labelings of conditions A and B across the n subjects, and each occurs with equal probability under H_0 .

Permutation Distribution: Calculate $\{T_v^j\}_{j=1}^{2^n}$ and $\{T_{\max}^j\}_{j=1}^{2^n}$, where T_v^j and $T_{\max}^j$ denote the test statistics for the j -th relabeling of conditions A and B.

Permutation p-values: Compare observed T_v and $T_{\max}$ to the permutation distributions $\{T_v^j\}_{j=1}^{2^n}$ and $\{T_{\max}^j\}_{j=1}^{2^n}$ to obtain p-values.

Simulation Design

Manipulated two factors: the distribution for $\epsilon_{iv} = A_{iv} - B_{iv}$ (see Figure 2), and the sample size $n \in \{10, 25, 50, 100, 200\}$.

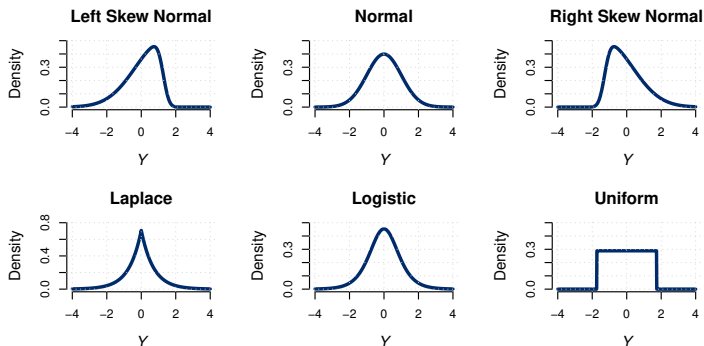


Figure 2: Six univariate distributions with mean $\mu = 0$ and variance $\sigma^2 = 1$.

Simulation Analyses

10,000 independent datasets (replications) were generated for each of the 30 ($6 \text{ distribution} \times 5 n$) cells of the simulation design.

For each replication, the null hypothesis $H_0 : \mu = 0$ was tested using the alternative hypothesis $H_1 : \mu > 0$ and an $\alpha = 0.05$ significance level.

The significance testing results were compared using two methods:

- (i) Student's t test
- (ii) Fisher's randomization test

The number of resamples was set at $R = \min(2^n, 9999)$.

Simulation Results

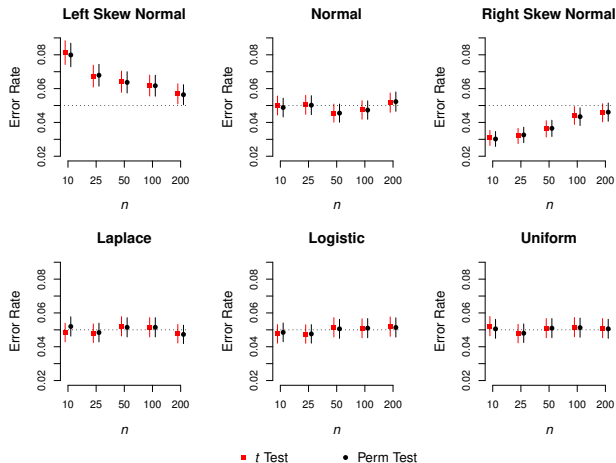


Figure 3: Type I error rate calculated across 10,000 replications for the t test (red squares) and the permutation test (black circles). The vertical bars denote approximate 99% confidence intervals, and the dotted horizontal lines denote the nominal rate of $\alpha = 0.05$.

Two Sample Location Tests

Motivating Problem

Let $G_i \in \{1, 2\}$ denote the group label for the i -th subject ($i = 1, \dots, n$), and suppose we want to test for group differences in the brain images.

Consider the following model separately at each voxel

$$Y_{iv} = \mu_{G_{iv}} + \epsilon_{iv} \quad (4)$$

where

- Y_{iv} is the i -th subject's recorded brain activity at the v -th voxel
- μ_{1v} and μ_{2v} are the unknown group mean parameters at the v -th voxel
- ϵ_{iv} are iid error terms with mean zero and variance σ_v^2

Want to test **local null hypothesis** separately at each voxel $H_0^v : \mu_{1v} = \mu_{2v}$ and the **global null hypothesis** across all V voxels $H_0 : (\forall v)(\mu_{1v} = \mu_{2v})$.

Test Statistics

For $H_0^v : \mu_{1v} = \mu_{2v}$, use the Student or Welch (1938, 1947) test statistic

$$T_v = \frac{\hat{\mu}_{1v} - \hat{\mu}_{2v}}{\hat{\sigma}_v \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad \text{or} \quad T_v^* = \frac{\hat{\mu}_{1v} - \hat{\mu}_{2v}}{\sqrt{\frac{\hat{\sigma}_{1v}^2}{n_1} + \frac{\hat{\sigma}_{2v}^2}{n_2}}} \quad (5)$$

where

- $\hat{\mu}_{kv} = \frac{1}{n_k} \sum_{i=1}^n \delta_{kG_i} Y_{iv}$ for $k \in \{1, 2\}$ (where δ_{kG_i} is Kronecker's delta)
- $\hat{\sigma}_v^2 = \frac{1}{n-2} \sum_{i=1}^n (Y_{iv} - \hat{\mu}_{G_iv})^2$ is the common error variance estimate
- $\hat{\sigma}_{kv}^2 = \frac{1}{n_k-1} \sum_{i=1}^n \delta_{kG_i} (Y_{iv} - \hat{\mu}_{G_iv})^2$ for $k \in \{1, 2\}$

For $H_0 : (\forall v)(\mu_{1v} = \mu_{2v})$, use either $T_{\max} = \max_v T_v$ or $T_{\max}^* = \max_v T_v^*$.

Permutation Inference

Randomization Assumption: The group labels G_i are arbitrary H_0 , i.e., $(Y_i|G_i = 1) \stackrel{d}{=} (Y_i|G_i = 2)$ where $Y_i = (Y_{i1}, \dots, Y_{iV})$.

Permutation Lemma: There are $M = \frac{n!}{n_1!n_2!}$ possible labelings of groups 1 and 2 across the n subjects, and each occurs with equal probability under H_0 .

Permutation Distribution: Calculate $\{T_v^j\}_{j=1}^M$ and $\{T_{\max}^j\}_{j=1}^M$, where T_v^j and $T_{\max}^j$ denote the test statistics for the j -th relabeling of subject groups.

Permutation p-values: Compare observed T_v and $T_{\max}$ to the permutation distributions $\{T_v^j\}_{j=1}^M$ and $\{T_{\max}^j\}_{j=1}^M$ to obtain p-values.

Simulation Design

Manipulated three factors:

- (i) the error distribution for ϵ_{iv} (see top row of Figure 2)
- (ii) the standard deviation of the second group: $\sigma_2 \in \{0.5, 1, 2\}$
- (iii) the sample size of the first group: $n_1 \in \{10, 25, 50, 100, 200\}$

Throughout the simulation...

- (a) the standard deviation of the first group was fixed at $\sigma_1 = 1$
- (b) the sample size of the second group was defined as $n_2 = 2n_1$

Simulation Analyses

10,000 independent datasets (replications) were generated for each of the 45 ($3 \text{ distribution} \times 3 \sigma_2 \times 5 n_1$) cells of the simulation design.

For each replication, the null hypothesis $H_0 : \mu_1 = \mu_2$ was tested using the alternative hypothesis $H_1 : \mu_1 > \mu_2$ and an $\alpha = 0.05$ significance level.

The significance testing results were compared using four methods:

- (i) Student's t test which assumes normality and equal variances
- (ii) Welch's t test which assumes normality and unequal variances
- (iii) the permutation test using Student's (pooled variance) T test statistic
- (iv) the permutation test using Welch's (unequal variance) T^* test statistic.

The number of resamples was set at $R = 9999$.

Simulation Results

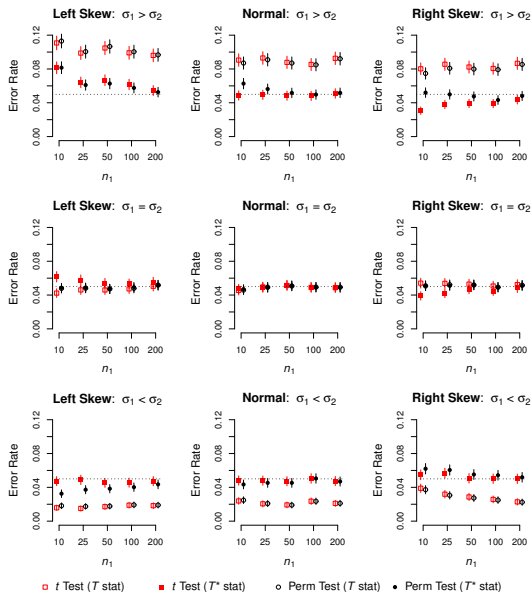


Figure 4: Type I error rate calculated across 10,000 replications for the t test (red squares) and the permutation test (black circles). The unfilled points denote results using the (Student) T test statistic, whereas the filled points denote results using the (Welch) T^* test statistic. The vertical bars denote approximate 99% confidence intervals, and the dotted horizontal lines denote the nominal rate of $\alpha = 0.05$.

Correlation Tests

Motivating Problem

Suppose that Y_{iv} is the recorded brain signal at the v -th voxel ($v = 1, \dots, V$) from the i -th subject ($i = 1, \dots, n$), and X_i is some continuous variable.

Consider the (Pearson product-moment) correlation separately at each voxel

$$\rho_{XY_v} = \frac{\sigma_{XY_v}}{\sigma_X \sigma_{Y_v}} \quad (6)$$

where

- $\mu_X = E[X]$ and $\mu_{Y_v} = E[Y_v]$ are the expectations
- $\sigma_X^2 = E[(X - \mu_X)^2]$ and $\sigma_{Y_v}^2 = E[(Y_v - \mu_{Y_v})^2]$ are the variances
- $\sigma_{XY_v} = E[(X - \mu_X)(Y_v - \mu_{Y_v})]$ is the covariance between X and Y_v

Want to test **local null hypothesis** separately at each voxel $H_0^v : \rho(X, Y_v) = 0$ and the **global null hypothesis** across all V voxels $H_0 : (\forall v)(\rho(X, Y_v) = 0)$.

Test Statistics

For $H_0^v : \rho(X, Y_v) = 0$, use the test statistic

$$T_v = \sqrt{n} \hat{\rho}_{XY_v} = \sqrt{n} \frac{\hat{\sigma}_{XY_v}}{\hat{\sigma}_X \hat{\sigma}_{Y_v}} \quad \text{or} \quad T_v^* = \frac{\hat{\rho}_{XY_v}}{\hat{\tau}_v / \sqrt{n}} \quad (7)$$

where

- $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ and $\bar{Y}_v = \frac{1}{n} \sum_{i=1}^n Y_{iv}$ are the means
- $\hat{\sigma}_X^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$ and $\hat{\sigma}_{Y_v}^2 = \frac{1}{n} \sum_{i=1}^n (Y_{iv} - \bar{Y}_v)^2$ are the variances
- $\hat{\sigma}_{XY_v} = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})(Y_{iv} - \bar{Y}_v)$ is the covariance
- $\hat{\tau}_v^2 = \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 (Y_{iv} - \bar{Y}_v)^2}{\hat{\sigma}_X^2 \hat{\sigma}_{Y_v}^2}$ is the asymptotic variance of T_v

For $H_0 : (\forall v)(\rho(X, Y_v) = 0)$, use either $T_{\max} = \max_v T_v$ or $T_{\max}^* = \max_v T_v^*$.

Permutation Inference

Randomization Assumption: The X and Y_v variables are independent for all v , i.e., $F_{XY_v} = F_X F_{Y_v}$ where $\{F_{XY_v}, F_X, F_{Y_v}\}$ are joint and marginal distributions.

Permutation Lemma: There are $n!$ possible pairings of the X_i 's with the Y_{iv} 's, and each occurs with equal probability under H_0 .

Permutation Distribution: Calculate $\{T_v^j\}_{j=1}^{n!}$ and $\{T_{\max}^j\}_{j=1}^{n!}$, where T_v^j and $T_{\max}^j$ denote the test statistics for the j -th permutation of the X_i 's.

Permutation p-values: Compare observed T_v and $T_{\max}$ to the permutation distributions $\{T_v^j\}_{j=1}^{n!}$ and $\{T_{\max}^j\}_{j=1}^{n!}$ to obtain p-values.

Simulation Design

Manipulated two factors: the data generating distribution (see Figure 5), and the sample size $n \in \{10, 25, 50, 100, 200\}$

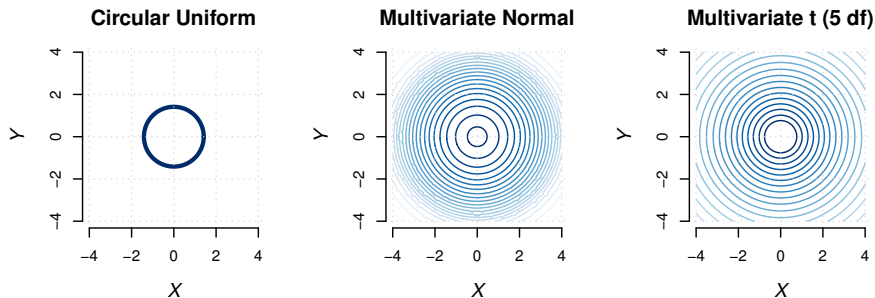


Figure 5: Three bivariate distributions with $\mu_X = \mu_Y = 0$, $\sigma_X^2 = \sigma_Y^2 = 1$, and $\rho_{XY} = 0$.

Simulation Analyses

10,000 independent datasets (replications) were generated for each of the 15 (3 distribution $\times$ 5 n) cells of the simulation design.

For each replication, the null hypothesis $H_0 : \rho(X, Y) = 0$ was tested using the alternative hypothesis $H_0 : \rho(X, Y) > 0$ and an $\alpha = 0.05$ significance level.

The significance testing results were compared using four methods:

- (i) the classic t test assuming bivariate normality
- (ii) a normal approximation using a studentized t test statistic
- (iii) the classic Pitman permutation test
- (iv) a studentized version of the classic permutation test

The number of resamples was set at $R = 9999$.

Simulation Results

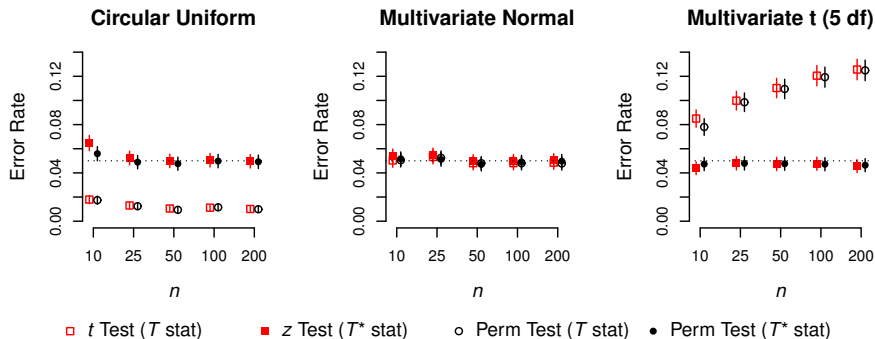


Figure 6: Type I error rate calculated across 10,000 replications for the parametric tests (red squares) and the permutation tests (black circles). The unfilled points denote the results using the (classic) T test statistic, whereas the filled points denote the results using the (studentized) T^* test statistic. The vertical bars denote approximate 99% confidence intervals, and the dotted horizontal lines denote the nominal rate of $\alpha = 0.05$.

Regression Tests

Motivating Problem

Consider the linear model separately at each voxel

$$Y_{iv} = \alpha_v + \mathbf{X}_i^\top \boldsymbol{\beta}_v + \mathbf{Z}_i^\top \boldsymbol{\gamma}_v + \epsilon_{iv} \quad (8)$$

where

- Y_{iv} is the response recorded from the i -th subject at the v -th voxel
- $\mathbf{X}_i = (X_{i1}, \dots, X_{ip})^\top$ and $\mathbf{Z}_i = (Z_{i1}, \dots, Z_{iq})^\top$ are predictor vectors
- $\alpha_v, \boldsymbol{\beta}_v = (\beta_{1v}, \dots, \beta_{pv})^\top$, and $\boldsymbol{\gamma}_v = (\gamma_{1v}, \dots, \gamma_{qv})^\top$ are coefficients
- ϵ_{iv} is an error term satisfying $E(\epsilon_{iv}) = 0$ and $E(\epsilon_{iv} | \mathbf{X}_i, \mathbf{Z}_i) = 0$

Want to test **local null hypothesis** separately at each voxel $H_0^v : \boldsymbol{\beta}_v = \mathbf{0}$ and the **global null hypothesis** across all V voxels $H_0 : (\forall v)(\boldsymbol{\beta}_v = \mathbf{0})$.

Test Statistics

For $H_0^v : \beta_v = \mathbf{0}$, use the test statistic

$$F_v = \frac{\hat{\theta}_v^\top \mathbf{R} [\mathbf{R}^\top (\mathbf{M}_c^\top \mathbf{M}_c)^{-1} \mathbf{R}]^{-1} \mathbf{R}^\top \hat{\theta}_v}{p \hat{\sigma}_v^2} \quad \text{or} \quad W_v = n \hat{\theta}_v^\top \mathbf{R} \left(\mathbf{R}^\top \hat{\Sigma}_*^{-1} \hat{\Omega}_v^* \hat{\Sigma}_*^{-1} \mathbf{R} \right)^{-1} \mathbf{R}^\top \hat{\theta}_v \quad (9)$$

where

- $\mathbf{R}$ is an $r \times p$ matrix such that $\mathbf{R}^\top \theta_v = \beta_v$ with $\theta_v = (\beta_v^\top, \gamma_v^\top)^\top$,
- $\mathbf{M}_c = [\mathbf{X}_c, \mathbf{Z}_c]$ is the columnwise mean-centered design matrix
- $\hat{\sigma}_v^2 = \frac{1}{n-r-1} \sum_{i=1}^n \hat{\epsilon}_{iv}^2$ with $\hat{\epsilon}_{iv} = Y_{iv} - \hat{Y}_{iv}$ and $\hat{Y}_{iv} = \hat{\alpha}_v + \mathbf{X}_i^\top \hat{\beta}_v + \mathbf{Z}_i^\top \hat{\gamma}_v$
- $\hat{\Sigma}_* = \frac{1}{n} \mathbf{M}_c' \mathbf{M}_c$ and $\hat{\Omega}_v^* = \frac{1}{n} \mathbf{M}_c' \mathbf{D}_v^{\hat{\epsilon}} \mathbf{M}_c$ where $\mathbf{D}_v^{\hat{\epsilon}} = \text{diag}(\hat{\epsilon}_{1v}, \dots, \hat{\epsilon}_{nv})$

For $H_0 : (\forall v)(\beta_v = \mathbf{0})$, use either $F_{\max} = \max_v F_v$ or $W_{\max} = \max_v W_v$.

Permutation Inference

Randomization Assumption: The $\mathbf{X}$ and $(\mathbf{Z}, Y_v)$ variables are independent $\forall v$, i.e., $F_{XZY_v} = F_X F_{ZY_v}$ where $\{F_{XZY_v}, F_X, F_{ZY_v}\}$ are joint and marginals.

Permutation Lemma: There are $n!$ possible pairings of the X_i 's with the Y_{iv} 's, and each occurs with equal probability under H_0 .

Permutation Distribution: Calculate $\{F_v^j\}_{j=1}^{n!}$ and $\{F_{\max}^j\}_{j=1}^{n!}$, where F_v^j and $F_{\max}^j$ denote the test statistics for the j -th permutation of the X_i 's.

Permutation p-values: Compare observed F_v and $F_{\max}$ to the permutation distributions $\{F_v^j\}_{j=1}^{n!}$ and $\{F_{\max}^j\}_{j=1}^{n!}$ to obtain p-values.

Simulation Design

Manipulated two factors: the data generating distribution (see Figure 7), and the sample size $n \in \{10, 25, 50, 100, 200\}$.

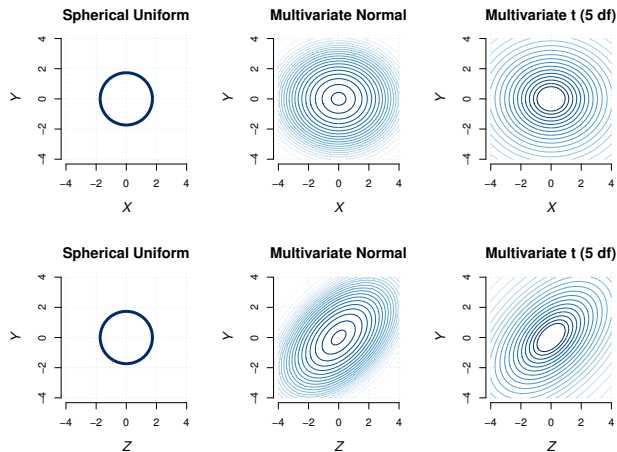


Figure 7: The X-Y slice with $Z = 0$ (top) and the Z-Y slice with $X = 0$ (bottom) for three trivariate distributions with $\mu_j = 0$ and $\sigma_j^2 = 1$ for $j \in \{X, Y, Z\}$. The X-Z slice with $Y = 0$ is identical to the top row. Note that $\rho_{XZ} = \rho_{XY} = 0$ for all three distributions, $\rho_{ZY} = 0$ for the spherical uniform, and $\rho_{ZY} = 1/2$ for the multivariate normal and multivariate t .

Simulation Analyses

10,000 independent datasets (replications) were generated for each of the 15 (3 distribution $\times$ 5 n) cells of the simulation design.

For each replication, the null hypothesis $H_0 : \beta = 0$ was tested using the alternative hypothesis $H_0 : \beta \neq 0$ and an $\alpha = 0.05$ significance level.

The significance testing results were compared using four methods:

- (i) the classic F test which assumes normality
- (ii) the classic Wald test which assumes a large sample
- (iii) a nonparametric permutation test using the F test statistic
- (iv) a nonparametric permutation test using the W (Wald) test statistic

The number of resamples was set at $R = 9999$.

Simulation Results

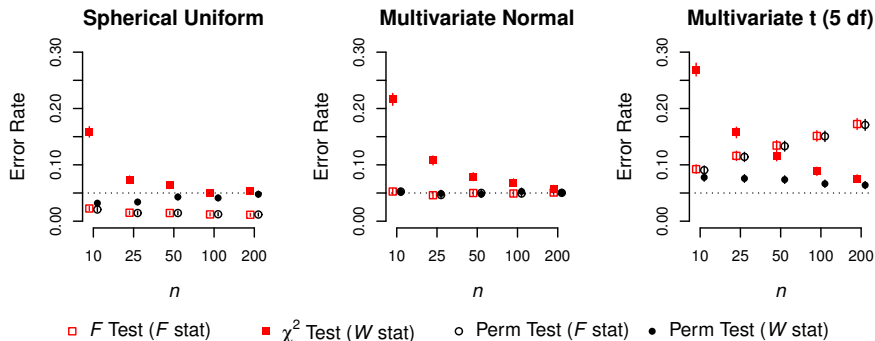


Figure 8: Type I error rate calculated across 10,000 replications for the parametric tests (red squares) and the permutation tests (black circles). The unfilled points denote the results using the (classic) F test statistic, whereas the filled points denote the results using the (Wald) W test statistic. The vertical bars denote approximate 99% confidence intervals, and the dotted horizontal lines denote the nominal rate of $\alpha = 0.05$.

Conclusions and Suggestions

Recap of Main Points

Permutation tests are becoming more popular in neuroimaging research.

- First proposed for one and two sample locations tests
- Now used for more complicated linear models w/ nuisance parameters

There are *several* important decisions that one must make when conducting a permutation test of neuroimaging data.

These decisions have important consequences for valid statistical inference.

Current implementations of permutation tests in neuroimaging are *not* robust to modest violations of the model assumptions. . . But they could be!

Permutation versus Sample Distributions

One of the key points of confusion in the neuroimaging literature seems to be the distinction (or lack thereof) between a permutation distribution and a sampling distribution of a test statistic.

A permutation distribution is not equivalent to the true sampling distribution of test statistic; however, if certain assumptions are met, a permutation distribution can provide a reasonable approximation to a test statistic's sampling distribution.

Understanding this distinction is essential for correctly interpreting the results obtained from a permutation test.

Clarifying Some Claims

When discussing permutation tests, Fisher (1960) interestingly noted that “Somewhat extravagant claims have often been made on their behalf”, see Chapter III (Section 21.1, “*Non-parametric*” Tests).

Claim 1 Permutation tests are flexible with respect to the chosen test statistic

- True, but the test statistic will affect the conclusions that can be drawn
- One needs to carefully consider the relationship between the test statistic, the null hypothesis, and the permutation procedure

Claim 2 Permutation tests are a good idea for small samples

- Only true if the sample is well-behaved, i.e., meets the assumptions
- For small samples, test statistics have no nice (asymptotic) properties

Moving Forward...

As computing resources improve, nonparametric permutation tests will continue to grow in popularity in the field of neuroimaging.

Nonparametric tests have the appeal of requiring less stringent assumptions than their parametric counterparts.

These ‘nonparametric’ approaches still do require some basic assumptions to provide valid statistical inference.

By incorporating recent theoretical advances in statistics, common implementations of permutation tests in neuroimaging could be improved.

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